

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
NEW COURSE

Exam.	Regular (Model Question)		
Level	BE	Full Marks	30
Programme	BCE	Pass Marks	12
Year / Part	II/ II	Time	1.5 hrs.

Subject: - Building Technology

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ **Necessary figures are attached herewith.**
- ✓ Assume suitable data if necessary.

1. Define built environment. Explain the history of building technology. [2]
2. Explain daylight factor and natural ventilation due to wind effects. How passive control system is achieved in building? Provide examples. [2+3]
3. a) What special thing should be kept in mind while excavating on sloppy land? Where and why strap footing is used in building? [2+2]
b) What are the different tools used for plastering? Differentiate construction joint and seismic joints in building. [2]
4. a) Explain the different terminology used in wooden doors. What is the advantage of suspended floor over solid floor. [2+2]
b) Describe lift, escalator and ramps. What is the different sanitary fitting used in building? [1.5+1.5]
5. a) Define and classify sound absorbents? Explain various methods of sound insulation through walls of a hall building. [2+2]
b) What are the different underpinning methods? Explain steps to be taken before demolition. [1+2]
6. Explain the key principles of sustainable building. [3]

Tribhuvan University
Institute of Engineering
Examination Control Division
NEW COURSE

Exam	Regular (Model Question)		
Level	BE	F.M.	60
Program	BEI	P.M.	24
Year/ Part	II/ II	Time	3 Hrs

Subject: Computer Organization and Architecture (BNEX 253)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
 - ✓ Attempt all questions
 - ✓ The figures in the margin indicate **Full Marks**.
 - ✓ Assume suitable data if necessary
1. What do you mean by overlapped register window? Explain the computer function with example. (4+4)
 2. Explain the Data transfer instruction and Data manipulation instruction with example. (8)
 3. Differentiate between microinstruction and micro operation. Explain the Microinstruction formats with example. (2+8)
 4. What do you mean by memory hierarchy? Explain in detail the different mapping function used in cache memory. (2+6)
 5. Write an algorithm and flowchart of non-restoring division algorithm. Divide 12/5 using non-restoring division algorithm. (5+5)
 6. Differentiate between arithmetic pipeline and instruction pipeline. (5)
 7. Explain how DMA controller is initialized for DMA data transfer? (5)
 8. Define multicore computer. Explain the increase in parallelism in multicore computer. (2+4)

MODEL QUESTION

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Exam Control Division
2082,

Exam Level
Program:
Year/ Part:

BE
BCE/BEL
II/II

Full Marks: 60
Pass Marks: 24
Time: 3 hrs.

Subject: Communication English (SH 204)

Candidates must give their answers in their own words as far as practicable.

Attempt ALL questions.

The figures in the margin indicate Full Marks.

1. Answer ANY ONE of the following questions [5 Marks]
 - a) Define paraphrasing plagiarism and patchwork plagiarism.
 - b) What is a copyright? How does it protect creators?
2. Answer ANY TWO of the following questions. Question no. 'C' is compulsory.
[2*5=10 Marks]
 - (a) Write bias-free words of the following:
foreman, delivery boy, repairman, serviceman, corporate wife, middleman, freshman,
forefathers, fatherland, rise of man
 - (b) What are the principles of effective technical writing?
 - (c) Rewrite the following sentences with correct answers:
 - i.) There (seem) to be three deer hiding in the bush at present.
 - ii.) Shanti's dog who is my neighbour has gone mad. (Edit the sentence)
 - iii.) The poet sat in this tower, and gazed on the sea. (Change into a simple sentence)
 - iv.) Notwithstanding several efforts, he failed. (Change into a compound sentence)
 - v.) You have earned his gratitude, so you shall not go unrewarded. (Change into a complex sentence.)
 - vi.) Nothing but cats and dogs (be) seen there now.
 - vii.) He was the most eloquent speaker.....I ever heard. (Use pronoun.)
 - viii.) With a view to (help) him, she came early.
 - ix.) Rather than (go) out, she started doing her homework.
 - x.) To err is human to forgive divine. (Use punctuation.)
3. Answer ANY THREE of the following questions. Question numbers 'a' and 'b' are compulsory.
[7+7+6 = 20 Marks]
 - a) Write a report on "Controlling Sound Pollution in Kathmandu". Mention only the main section of your report.
 - b) A large public limited company has been spending about rupees seven lakhs per annum on getting its various documents such as routine forms, brochures, reports, proposals, manuals, etc, printed by outside agencies. For greater economy and efficiency it has decided to have its own Publishing Unit to cater to all its printing needs. As Office Manager write a proposal for the creation of this unit for consideration of the Board of Directors. Prepare title page, abstract, and cost estimate parts of the proposal.
 - c) Write an abstract, the statement of problem, research questions and findings of a research article titled "The Ecological Impact of Using Digital Technology".
 - d) Define in-text citation and prepare a reference page of the given information in APA style of citation:

i.) Name of authors: B.B. Karki

Name of journal: Journal of Nepalese Business Studies

Title of article: Doing Business and Role of Government for Entrepreneurship Development

Volume and issue numbers: 7(1)

Year of publication: 2010

Page numbers: 53-62

ii.) Name of authors: Meenakshi Raman and Sangeeta Sharma

Name of Book: Technical Communication: Principles and Practice

Year of publication: 2022

Edition: 4th

Place of publication: New Delhi

Name of publisher: Oxford University Press

4. Answer ANY TWO of the following questions: [2*5=10 Marks]
- a) You have a Bachelor's in Computer Engineering degree. Now you want to get admission to the Master of Computer Engineering program at one of the best universities in America. Inventing details, write a letter asking for the necessary information.
 - b) Your college is organizing a one-day workshop on "Agricultural Entrepreneurship and Youth Employment". Write a press release inviting local media to cover the event.
 - c) Suppose that you are the Manager of a bank. You noticed that your staff are not using electricity, office stationery and other equipment prudently. Write them a Memo asking them not to make a haphazard use of office gadgets.
5. Answer ANY TWO of the following questions: [2*5=10 Marks]
- a) List the fundamentals of effective speaking. Discuss the role of pace in effective speaking.
 - b) How can you overcome fear and nervousness of public speaking?
 - c) What is active listening? Describe any two barriers to effective listening.
6. Answer ANY ONE of the following questions: [1*5=5 Marks]
- a) Create a bar graph to show the work schedule of any proposed work.
 - b) Differentiate between diagrams and tables.

Examination	Regular (Model Question)		
Level	BE/BAR	Full Marks	60
Program	BEL	Pass Marks	24
Year/Part	II/II	Time	3 Hrs.

Subject: - Control System

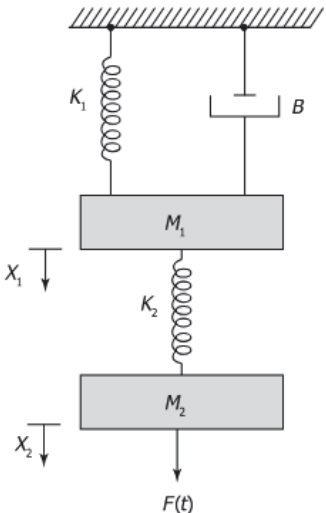
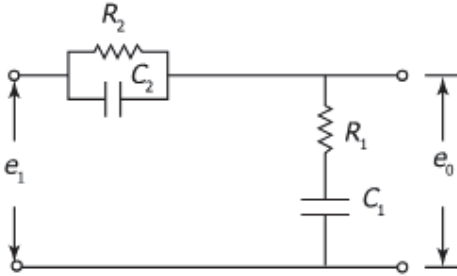
Candidates are required to give their answers in their own words as far as practicable.

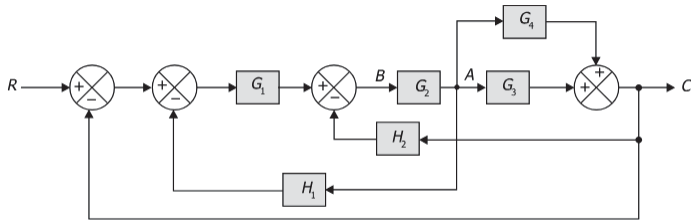
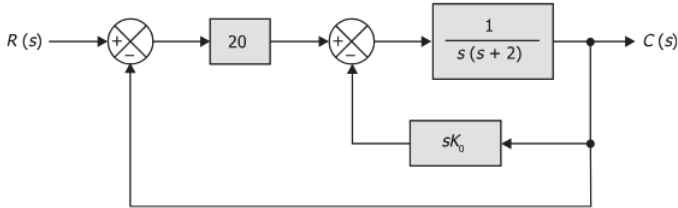
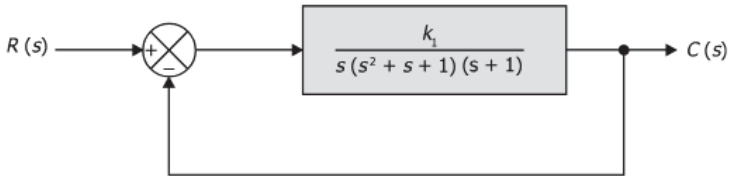
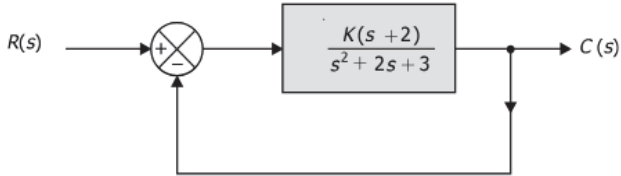
Attempt **All** questions

The figures in the margin indicate **Full Marks**

Necessary figure(s) are/is attached herewith.

Assume suitable data if necessary

1.	How you define systems? Explain with examples of systems with their description you encounter in your daily life.	1+1
2. a	What is mathematical modelling? The figure below shows, obtain the transfer function $\frac{X_1(s)}{F(s)}$. 	2+6
2.b.	Obtain the transfer function $\frac{v_0(s)}{v_s(s)}$ of the following electric filter compensator. 	6
3.a.		6

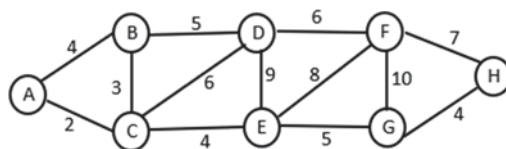
	Reduce the following block diagram representation of a system using reduction algebra to obtain $\frac{Y(s)}{R(s)}$. 	
3.b.	A feedback system having derivative feedback control has been shown in figure below:  i) If $K_0 = 0$, determine the damping ratio and natural frequency of the system. What is the steady-state error to a unit ramp input? ii) Determine the value of K_0 for which the damping ratio of the system will be 0.6. Find the steady state error to a unit ramp input for this value of K_0.	3+3
4.a	Find the range of K for system stability for the following system. 	3
4.b	Mentioning the significance of root locus on control system design, Sketch the root locus for the following closed loop system and obtain the value of K for which system becomes critically damped and point out in the graph. 	5
5.a.	Explain how frequency response technique can be used to determine the stability of closed loop feedback system. Using Nyquist Stability Criterion, determine the closed loop stability of system having open loop transfer function given as: $G(s).H(s) = \frac{s+2}{s-1}$	6

5.b.	Obtain the phase margin of the system having transfer function: $G(s) = \frac{40}{s(s + 8)}$	5
6.a	A control system is represented by open loop transfer function given as: $G(s) = \frac{K}{s(s + 3)}$	8
6.b.	It is desired that the damping ratio of the dominant roots of the system to be $\zeta = 0.5$ and settling time to be 2 sec. Additionally, the velocity error constant to be 10. Desing an appropriate compensator. Why state space representation became necessary in control systems? Obtain the state space representation of following transfer function. $G(s) = \frac{C(s)}{R(s)} = \frac{s + 3}{(s + 1)(s + 2)}$	2+3

Exam.	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BEI/BCT	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: Discrete Structure (ENCT 251)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
1. Define an equivalence relation with an example. Using truth table show that two compound propositions are logically equivalent. [2+2]
 2. Explain rules of inference for propositional logic. Show that the premises “There is someone in this class who has been to Tansen. Everyone who goes to Tansen visit the Shreenagar lead to the conclusion someone in this class has visited Shreenagar” using rules of inference [3+3]
 3. Explain vacuous proof method with an example. Explain the process to check the validity of an argument using method of tableaux. [2+2]
 4. Using mathematical induction show that $5^n - 1$ is divisible by 4 for $n \geq 1$. [3]
 5. Explain different representation of DFA with an example. [3]
 6. Write down the regular expression for the language in which the string contains the 15th symbol from the left as ‘a’ over an alphabet {a,b}. Show that regular language are closed under union and concatenation. [2+4]
 7. Define CFG. Construct a NFA for the following grammar and convert this NFA to its equivalent DFA. $G = \{N, T, P, S\}$ where $N = \text{set of non-terminals} = \{S, A, B\}$, $T = \text{set of terminals} = \{a, b\}$, $P = \text{set of productions} = \{S \rightarrow aA | bB, A \rightarrow a, B \rightarrow a\}$ and $S = \text{starting symbol}$ [1+4]
 8. Define linear non- homogeneous recurrence relation? Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 3^n$ with initial condition $a_1 = 5$. [4]
 9. What is an asymptotic notation ? Compute the best case and worst-case tie complexity for the insertion sort algorithm. [2+2]
 10. Define degree sequence of a graph. Explain Euler Circuit and Hamiltonian Circuit with example. [1+2+2]
 11. What is chromatic number? Find the minimum spanning tree for graph given below. [1+4]



12. Draw $K_{3,4}$ and write its adjacency matrix and incidence matrix. Explain different tree traversal methods. [3+3]
13. How shortest path can be obtained from source vertex to all other vertex using Dijkstra's algorithm? Explain with an example. [4]

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NEW COURSE

Exam.	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BCT	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: - Data Structure and Algorithm [ENCT 252]

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1.	Differentiate between primitive and non- primitive data types. Explain Asymptotic notation (big-oh, big-Omega, theta) with figure and example.	[2+3]
2.	Convert the following infix express notation into postfix. $(A/B)*(C*F+(A-D)*E)$. Evaluate resultant postfix expression if A=1, B=2, C=3, D=4, E=5, F=6. Draw a recursive tree for n=3 disks of Tower of Hanoi.	[4+1+3]
3.	Differentiate between linear queue and double-ended queue. Write algorithms for enqueue, dequeue and traverse operations in an array-based circular queue. How does circular queue overcome the limitations of linear queue?	[2+4+1]
4.	Explain static implementation of a list, how it differs from dynamic implementation? Write an algorithm to insert node at the end of singly linked list.	[3+4]
5.	Define BST. Write down the algorithm for inserting a new node in BST. Calculate the percentage of compression for following characters using Huffman Code. AAATTTTTTTEEEEEEEEEEOOOOO	[1+4+4]
6.	Explain Depth First traversal with an example. Explain Warshall's algorithm with suitable example.	[3+5]
7.	Define sorting. Classify different sorting algorithms based on stability and adaptiveness. Trace the steps of bubble sort for the array: [45, 23, 11, 89, 77, 98, 4]	[1+3+4]
8.	Explain binary search algorithm with suitable example. Define hashing. Draw the hash table obtained from double hashing with hash function: $h_1(k) = k \bmod 11$ and $h_2(k) = k \bmod 9$ to the keys given by: 10, 22, 31, 4, 15, 28, 17, 88 and 59 with table size m=11. Use $h(k,i) = (h_1(k) + i * h_2(k)) \bmod m$.	[3+1+4]

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Examination	Regular (Model Question)		
Level	BE	Full Marks	30
Program	BEL	Pass Marks	12
Year/Part	II/II	Time	3 hrs.

Subject: - Electric Machines II (ENEE253)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicates full marks.
- ✓ Assume suitable data if necessary.

1. Compare rotating armature and stationary armature constructions in synchronous machines. [3]
Which is more suitable for large machines and why?
2. A 3-phase star-connected alternator is rated at 1,600 kVA, 13.5 kV. The per phase armature effective resistance and synchronous reactance are 1.5 ohm and 30 ohm respectively. Calculate voltage regulation for a load of 1.280 MW at power factors of (i) 0.8 leading (ii) unity and (iii) 0.8 lagging. [3]
3. Explain in details about the parallel operation of two synchronous generator by speed droop characteristics. [3]
4. With the help of relevant curves and diagram, explain the ampere turn method for determining the voltage regulation of a synchronous generator [3]
5. Derive the power angle characteristics of salient pole synchronous machine and also derive condition for maximum power. [3]
6. A salient pole synchronous motor has $X_d = 0.85$ pu and $X_q = 0.55$ pu. It is connected to bus-bars of 1.0 pu voltage, while its excitation is adjusted to 1.2 pu. Calculate the maximum power output, the motor can supply without loss of synchronism. Compute the minimum pu excitation that is necessary for the machine to stay in synchronism while supplying the full-load torque [3]
7. What is Park's Transformation. [3]
8. A single phase induction motor has $R_1 = 2 \Omega$, $X_1 = X_2' = 3.1 \Omega$, $R_2' = 1.98 \Omega$ and $X_{mag} = 40.17 \Omega$. If the motor is supplied from 240V single phase ac supply, Determine input current and power factor of the motor. [3]
9. Write working principal, advantages, disadvantages and application of reluctance motor. [3]
10. Write short note (Any Two) [3]
 - a) Split phase induction motor.
 - b) Capacitor start motor.
 - c) Capacitor start capacitor run motor.

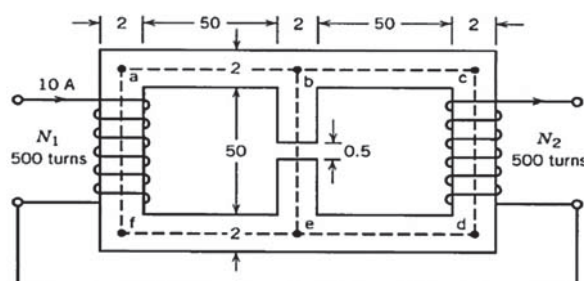
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Examination Control Division
NEW COURSE

Exam.	Regular (Model question)		
Level	BE	Full Marks	60
Programme	BME, BAME, BIE, BAS	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: - Electric Machines (EE.....)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt all questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

- For the given magnetic circuit, neglecting magnetic leakage and fringing, calculate air gap flux, flux density and magnetic field intensity at the air gap. Given that relative permeability of ferromagnetic material is 1200 and all dimensions are in cm. [5]



- Explain the construction and operation of single phase transformer. [5]
- A 80 kVA , $2000/200\text{ V}$, 50 Hz single phase transformer has impedance drop of 8% and resistance drop of 4% . Calculate the voltage regulation of transformer at full load at 0.8 pf lagging. Also find the power factor at which voltage regulation will be zero. [5]
- What is excitation in dc generator? Classify with equivalent circuit dc generators. [5]
- A dc series motor connected to 440 V supply runs at 600 rpm when taking a current of 50 A . calculate the value of resistance which when inserted in series with motor will reduce the speed to 400 rpm , the gross torque then being half the previous value. The total series resistance of motor is 0.2 ohm . [5]
- Briefly explain with proper reason: {any two} [2*3]
 - Induction generator is used with wind turbine.
 - Need of star delta starter for induction motor.
 - DC series motor has high starting torque.
- Explain the production of rotating magnetic field in three phase induction motor. [4]
- A 4 pole, 50 Hz , 3 phase induction motor develops a starting torque of 60 N-m . The rotor winding has an impedance of $(0.8+j2)\text{ }\Omega$ per phase at stand still. Calculate the maximum torque and speed at maximum torque. [5]
- A three phase, $11,000\text{ V}$, star connected alternator has armature resistance of $1\text{ }\Omega/\text{phase}$ and synchronous reactance of $20\text{ }\Omega/\text{phase}$. Find the percentage change in terminal voltage for a load of 1500 kW at power factor of 0.8 lagging and 0.8 leading. [5]
- What is hunting in synchronous motor? Explain the operation of three phase synchronous motor. [5]
- Classify single phase induction motors. Explain the construction and operation of universal motors. [5]
- Explain the construction and operation of permanent magnet synchronous motor. [5]

Subject: - Electromagnetics (ENEX 254)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume that the **boldfaced** letter represents vector and $\mathbf{a}_{\text{subscript}}$ represents a unit vector.
- ✓ Assume suitable data if necessary.

1. Transform the vector $\mathbf{A} = y\mathbf{a}_x + (x+y)\mathbf{a}_z$ at $P(-2,6,3)$ in cylindrical system. [5]
2. Explain Conservative Field. Show that electric field intensity as the negative gradient of scalar electric potential. [2+4]
3. Let $\mathbf{D} = y^2z^3\mathbf{a}_x + 2xyz^3\mathbf{a}_y + 3xy^2z^2\mathbf{a}_z$ pC/m² in free space. [6]
 - (a) Find the total electric flux passing the surface $x=3$, $0 \leq y \leq 2$, $0 \leq z \leq 1$ in the direction away from the origin.
 - (b) Find $|\mathbf{E}|$ at $P(3,2,1)$.
 - (c) Find the total charge contained in an incremental sphere having radius of $2\mu\text{m}$ centered at $P(3,2,1)$.
4. Explain the significance of Boundary Conditions. State and prove Uniqueness Theorem. [2+6]
5. Differentiate between Scalar Magnetic Potential and Vector Magnetic Potential. Find the magnetic field intensity at point P due to an infinite filament carrying current I placed on z-axis by using Biot-Savart Law. [2+4]
6. Evaluate both sides of Stoke's Theorem for the field $\mathbf{H} = 6xy\mathbf{a}_x - 3y^2\mathbf{a}_y$ A/m and the rectangular path around the region $2 < x < 5$, $-1 < y < 1$, $z=0$. Let the positive direction of $d\mathbf{s}$ be $\mathbf{a}_z$. [6]
7. Select the value of k so that each of the following pairs of fields satisfies Maxwell's equations in a region where $\sigma = 0$ and $\rho_v = 0$. [6]
 - (a) $\mathbf{E} = (kx - 100t)\mathbf{a}_y$ V/m, $\mathbf{H} = (x + 20t)\mathbf{a}_z$ A/m if $\mu = 0.25\text{H/m}$ & $\epsilon = 0.01\text{F/m}$
 - (b) $\mathbf{D} = 5x\mathbf{a}_x - 2y\mathbf{a}_y + kza_z$ $\mu\text{C/m}^2$, $\mathbf{B} = 2\mathbf{a}_y$ mT, $\mu = \mu_0$, $\epsilon = \epsilon_0$
 - (c) $\mathbf{E} = 60\sin(10^6t)\sin(0.01z)\mathbf{a}_x$ V/m, $\mathbf{H} = 0.6\cos(10^6t)\cos(0.01z)\mathbf{a}_y$ A/m, $\mu = k$.
8. Define Input Intrinsic Impedance. Derive the expression for input intrinsic impedance using the concept of reflection of uniform plane wave. [2+4]
9. A uniform plane wave in free space is given by $\mathbf{E}_s = (200 \angle 30^\circ)e^{-j250z}\mathbf{a}_x$ V/m. Find: (a) β (b) ω (c) $\mathbf{H}_s$ (d) $|\mathbf{E}|$ at $z = 8\text{mm}$ & $t = 6\text{ps}$. [6]
10. A lossless transmission line with $Z_0 = 60\Omega$ is 400m long. It is terminated with load $Z_L = 400 + j80\Omega$ and is operated with frequency of 1MHz. Let $v_p = 0.8c$. Find: [2+1+2]
 - a) The reflection coefficient(Γ)
 - b) The standing wave ratio on the line
 - c) The input impedance(Z_{in})

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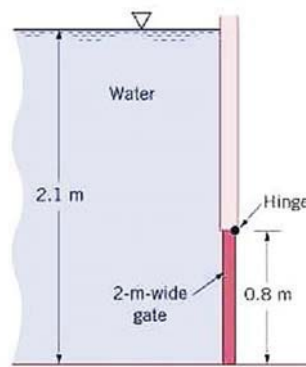
Exam	Regular (Model Question)		
Level	B.E.	Full Marks	60
Program	BIE/BAM	Pass Marks	24
Year/Part	II/II	Time	3 hrs

Subject: Fluid Mechanics and Machines

ENME 255

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **ALL** questions.
- ✓ Assume suitable data if necessary.

- 1 a) Two parallel plates kept 10 cm apart have laminar flow of oil between them with a maximum velocity of 1.5 m/s. If the viscosity of oil is 25 poise, then compute [6]
 - (i) the discharge per metre width,
 - (ii) maximum shear stress at the plates,
 - (iii) difference in pressure between two points 20 m apart,
 - (iv) velocity gradient at the plates and
 - (v) velocity at 2 cm from the plate. .
- b) Explain the working principle of Kaplan turbine with neat diagram and necessary equations. [6]
- 2 a) Determine the magnitude and direction of the force that must be supplied to the bottom of the gate to keep the gate closed. [6]



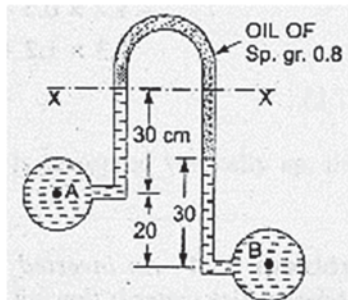
- b) What is meant by Buckingham Π theorem? Explain with necessities equations. [6]
3. a) State the assumptions and limitations of Bernoulli's equation. State continuity equation with its application in fluid flow. [6]
- b) The velocity components of the two dimensional plane motion of a fluid are expressed as $u = \frac{y^3}{3} + 2x - x^2y$; $v = xy^2 - 2y - \frac{x^3}{3}$. Show that these functions represent possible case of an irrotational flow. Obtain the expression for stream function. [6]
- 4 a) What is meant by adverse pressure gradient? Why flow separation is important in fluid mechanics? [6]
- b) What are various types of minor loss? How can moody diagram be used to calculate the head loss in pipe flow? [6]
5. a) What are various types of flow measuring devices? Explain any one with neat diagram. [6]
- b) Describe the working principle of a centrifugal pump with a labeled diagram. [6]

Exam	Regular (Model Question)		
Level	BE	Full Marks	60
Program	BME/BAS	Pass Marks	24
Year / Part	II / II	Time	3 Hrs.

Subject: - Fluid Mechanics with Engineering Applications

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ All questions carry equal marks.
- ✓ Assume suitable data if necessary.

1. A differential manometer is connected to two pipes A and B which convey water. The fluid in manometer is oil of sp. Gr. 0.8. For the manometer reading shown in the figure, find the pressure difference between A and B.



2. What are various types of flow measuring devices? Differentiate between rotameter and triangular notch in terms of flow measurement.
3. Use dimensional analysis to derive the expression for the time period of a simple pendulum.
4. Prove that for steady laminar flow between two fixed parallel plates, the velocity distribution across a section is parabolic and that the average velocity is $2/3^{\text{rd}}$ of the maximum velocity.
5. Using boundary layer theory, explain the velocity profile change before and after separation.
6. A cylindrical can 150 mm in diameter is filled to a depth of 100 mm with a fuel oil. The oil has a mass of 1.5 kg. Calculate its density, specific weight and specific gravity.
7. A pipe 300 m long has a slope of 1 in 100 and tapers from 1.2 m diameter at the high end to 0.6 m diameter at the low end. Quantity of water flowing is 5400 liters per minute. If the pressure at the high end is 68.67 kPa, find the pressure at the low end. Neglect losses.
8. What are various types of dimensionless number in fluid mechanics? Explain with necessary equations.
9. Consider the velocity field given by

$$u = \frac{y}{x^2 + y^2} \text{ and } v = -\frac{x}{x^2 + y^2}$$

Calculate the equation of streamline passing through the point (0,5)

10. Differentiate between major head loss and minor head loss. Explain the effect of doubling pipe diameter on head loss if flow rate remains constant.

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NEW COURSE

Examination	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BAR	Pass Marks	24
Year/Part	II/II	Time	3 Hrs.

Subject: - History of Western Architecture (*ENAR 253*)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
 - ✓ Attempt **All** questions.
 - ✓ The figures in the margin indicate **Full Marks**.
 - ✓ Assume suitable data if necessary.
-
1. Discuss in brief how did religious beliefs and practices influence the design development of ancient Egyptian cult temple architecture. (4+4)
 2. Compare and contrast distinguishing elements of Greek classical orders, using relevant sketches. (4+4+4)
 3. Analyze the evolution of ancient Roman civic architecture in terms of its types, concepts, and uses. (4+4+4)
 4. Discuss the philosophies and applications of any one prominent architect's contribution to the development of the Renaissance, using examples. (6+6)
 5. Write short notes on: (*Any four*) (4X4)
 - a. Stonehenge
 - b. Early Christian church design
 - c. Characteristics of Romanesque architecture
 - d. Rims Cathedral
 - e. Cristal Palace

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROLLER DIVISION
(NEW COURSE)

Examination	Regular (Model Question)		
Level	Bachelor	Full Marks	60
Programme	BIE	Pass Marks	24
Year/Part	II/II	Time	3 Hrs.

Subject: Industrial Economics (ENIE252)

Candidates are required to give their answers in their own words as far as practicable.

Attempt **all** Questions.

The figures in the margin indicate full marks.

All required data are given in the question. Assume suitable data if necessary.

1. Why is the study of cost structures essential for industrial engineers? Explain with examples how demand forecasting supports production and inventory decisions. **[2+4]**
2. a) Define direct and indirect costs. Give one example of how each affects project budgeting. **[3]**
b) How can price elasticity be used to optimize promotional strategies in industries with fluctuating demand? **[3]**
c) Explain how oligopoly market structure affects innovation, pricing, and consumer access compared to monopolistic competition. **[3]**
3. a) How does the interest rate environment affect capital investment decisions in heavy industries? **[3]**
b) What is the role of public-private partnerships (PPPs) in implementing fiscal policies for industrial growth? **[2]**
c) How do skill shortages and labor regulations shape industrial policy? Compare how export promotion zones (EPZs) and domestic protectionism influence trade. **[2+2]**
4. A capital investment of Rs. 900,000 is expected to generate Rs. 180,000 annually for 7 years. Salvage value at the end is Rs. 100,000. Calculate NPV at 10% and determine the Benefit-Cost Ratio (BCR). **[8]**
5. A firm must decide between two compressors:

	Compressor X	Compressor Y
Initial Cost	Rs. 250,000	Rs. 400,000
Annual O&M Cost	Rs. 60,000	Rs. 20,000
Life (years)	4	8
Salvage Value	Rs. 25,000	Rs. 80,000

Assuming a discount rate of **10%**,

- a) Compute the Annual Equivalent Cost (AEC) of both options. **[5]**
- b) Recommend the better alternative based on life-cycle costing. **[2]**
6. a) What is the role of cash flow analysis in financial planning for industrial projects? Describe how it differs from profit-based evaluation. **[5]**
b) A project is evaluated with the following estimates:
 - Investment: Rs. 750,000
 - Revenue/year: Rs. 400,000
 - Variable cost/year: Rs. 180,000
 - Fixed cost/year: Rs. 90,000
 - Project life: 5 years
 - Discount rate: 10%

You are required to:

- a. i) Calculate the NPV. [4]
- ii) Perform sensitivity analysis for the following scenarios:
- a. Variable cost increases by 10%
 - b. Revenue drops by 8%
 - c. Fixed cost increases by 12% [2*3=6]
7. a) How do gig economy platforms transform cost structures and workforce models in traditional manufacturing or services? [3]
- b) Analyze how circular economy strategies reduce environmental externalities. How does compliance with global environmental standards (e.g., ISO 14001) impact industrial competitiveness? [3]

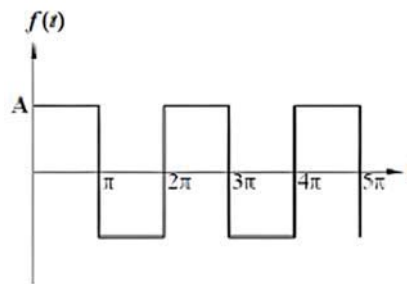
TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination and Control Division
(NEW COURSE)

Exam	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BME	Pass Marks	24
Year/Part	II/II	Time	3 hrs

Subject: Instrumentation and Sensor (ENME 252)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The Figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Differentiate between intrusive and non-intrusive techniques used in mechanical measurement systems. Why is calibration important in measurement system. [6+2]
2. Determine Fourier series expansion for the given time dependent signal as shown in [6]



3. Explain the working principle of strain gauge. Derive an expression for output voltage of a full-bridge circuit used for resistance to voltage conversion in mechanical measurement systems. [2+6]
4. A first order instrument has a time constant of 0.4 s. For a step input to the instrument, find the time at which the ratio of the output to the input signals is 0.9. [4]
5. A pressure transducer behaves as a second order system. If the undamped natural frequency is 4000 Hz and the damping ratio is 80% of critical, determine the frequency range over which the measurement error is not greater than 10 %. What will be the frequency range if the damping ratio is changed to 0.4? [6]
6. How do you measure torque using strain gauge. Explain the concept of six axis load cell. [6]
7. What is piezoelectric effect. Describe the application of this effect in a measuring instrument with examples. [6]
8. Explain the working of thermocouple along with its advantages and disadvantages. Briefly explain type R thermocouple. [4]
9. Define MEMS accelerometers working principle. Explain significance of intelligent sensor in industrial systems. [4+2]
10. Explain the construction and working of pitot tube used for flow measurement. What is the minimum number of static ports required in pitot tube? [5+1]

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
NEW COURSE**

Exam	Regular (Model Question)		
Level	BE	Full Marks	60
Program	BEI/BCT	Pass Mark	24
Year/Part	II/II	Time	3 hours

Subject: Instrumentation

ENEX 252

- ✓ Candidates are required to give answer in their own words as far as practicable
 - ✓ Attempt All Questions
 - ✓ Assume Suitable data if necessary
 - ✓ Figures in the margin indicate the full marks
- 1 Illustrate open loop system with a useful example. 3
 - 2 Express the terms Accuracy, Precision and Sensitivity with an example 2+2+2
 - 3
 - a. A capacitive transducer uses two quartz diaphragms of area 600 mm^2 separated by a distance of 2.5mm. A pressure of $8 \times 10^5 \text{ N/m}^2$, when applied to the top diaphragm causes a deflection of 0.5mm. The capacitance is $400 \times 10^{-12} \text{ F}$ when no pressure is applied to the diaphragm. Determine the value of capacitance after the application of $8 \times 10^5 \text{ N/m}^2$ 4
 - b. Draw workflow of sensor in typical system. List out various sensors according to the principle of operation. 2+2
 - 4
 - a. How can you interface ADC using interrupt? Explain with suitable example 5
 - b. What do you mean by ARM processor? Briefly discuss the pin function and ports of At Mega 328P microcontroller.' 1+4
 - c. Explain how you interface GPS module with AT Mega 328 microcontroller using UART communication Protocol 4
 - 5
 - a. Draw frame structure of UART communication Protocol. Explain Bluetooth protocol stack in wireless communication 1+3
 - b. Sketch the block diagram of the data logger labeling all required parameters. 2
 - 6
 - a. Define pseudo impedance. Explain Ground Loop mechanism and also discuss the elimination of the ground loop issues with suitable diagrams. 1+3
 - 7
 - a. Which software model is generally preferred for highly uncertain and early stage startups? Explain its working principle. 4
 - b. Differentiate black box testing and white box testing 2
 - 8 Describe the construction and working principle of a single phase induction type energy meter 6
 - 9 Define the term Industry 4.0. List out some characteristic of IoT system. 1+2
 - 10 How you interface any actuator with a microcontroller of your choice in an industrial instrumentation system. Draw a required interfacing circuit and explain its simple operation. 4

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
NEW COURSE

Examination	Regular (Model Question)		
Level	BE	Full Marks	60
Program	BEL	Pass Marks	24
Year/Part	II/II	Time	3 hrs.

Subject: Microprocessor and Microcontroller (ENEX 251)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**
- ✓ Assume suitable data if necessary.

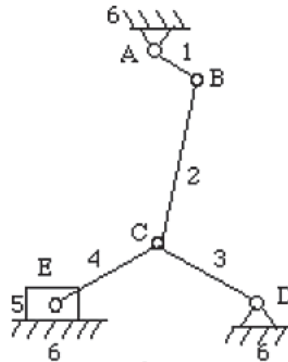
1. Draw a bus organization of microprocessor. Define Stored program concept. [2+2 = 4]
2. a) Define instruction and instruction cycle. Draw the block diagram of programing model of 8085 microprocessor [2+4 =6]
b) Write a program in 8085 that will check whether the bit D₆ is 0 and D₃ is 1 of a number stored at 9000H. If the condition satisfies display the number at PORT 42H. [6]
3. a) What do you mean by instruction pipelining? Explain the addressing modes of 8086 with example. [1+5=6]
b) Write a program for 8086 microprocessor to read a string and display each word in separate line in the center of the screen. [6]
4. a) Design an address decoding circuit to interface one 4KB RAM chip and one 2KB ROM chip with 8085 microprocessors consecutively starting from the memory location 9000H. [6]
b) Draw a timing diagram of instruction LDA 3050H. Calculate the time required to execute such instruction, if microprocessor is operating at 3MHz clock frequency. [6]
5. a) Draw internal architecture of 8051 microcontroller and explain briefly about Array processing and look up table in 8051 microcontrollers. [2+4 =6]
b) In 8051, what do you mean by special function registers? Write an assembly language on 8051 to read an 8-bit temperature in Celsius from Port 1 and to output the Fahrenheit temperature equivalent onto Port 2. [2+4=6]
6. Write short notes on **ANY TWO**: [4 x 2 =8]
 - a. ISR and IVT in 8086
 - b. Flag of 8085
 - c. Microprocessor vs Microcontroller

Examination	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BAM	Pass Marks	24
Year/ Part	II/II	Time	3 hrs.

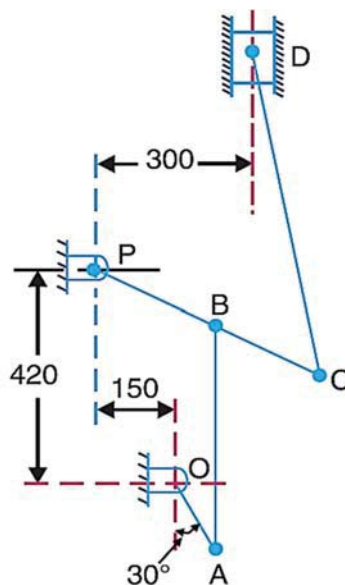
Subject: Theory of Machines and Mechanisms **Course Code:** ENME 256

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt ALL questions
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

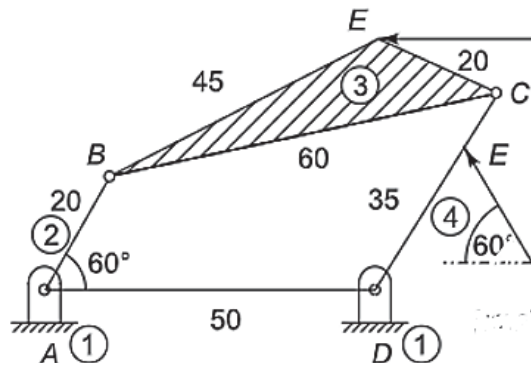
1. Calculate the degree of freedom of following figure.



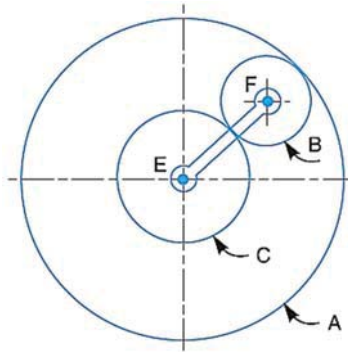
2. Find out the acceleration of the slider D and the angular acceleration of link CD for the engine mechanism shown in Figure. The crank OA rotates uniformly at 180 r.p.m. in clockwise direction. The various lengths are: OA = 150 mm ; AB = 450 mm; PB = 240 mm ; BC = 210 mm ; CD = 660 mm.



3. For a mechanism shown in figure below, determine the torque on link AB for static equilibrium of the mechanism. The dimensions of the various links are AB = 20 cm, triangle BCE with (BC = 60 cm, EC = 20 cm, and BE = 45 cm), DC = 35 cm, CF = 10cm, AD = 50cm. force on F=20N and force on E = 25N in direction as shown in figure.



4. A cam, with a minimum radius of 25 mm, rotating clockwise at a uniform speed is to be designed to give a roller follower, at the end of a valve rod, motion described below: 9
1. To raise the valve through 50 mm during 120° rotation of the cam;
 2. To keep the valve fully raised through next 30° ;
 3. To lower the valve during next 60° ; and
 4. To keep the valve closed during rest of the revolution i.e. 150° ;
- The diameter of the roller is 20 mm and the diameter of the cam shaft is 25 mm. Draw the profile of the cam when (a) the line of stroke of the valve rod passes through the axis of the cam shaft, and (b) the line of the stroke is offset 15 mm from the axis of the cam shaft. The displacement of the valve, while being raised and lowered, is to take place with simple harmonic motion. Determine the maximum acceleration of the valve rod when the cam shaft rotates at 100 r.p.m. Draw the displacement, the velocity and the acceleration diagrams for one complete revolution of the cam.
5. An epicyclic gear consists of three gears A, B and C as shown in Figure. The gear A has 72 9
- internal teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the center of A at 18 r.p.m. If the gear A is fixed, determine the speed of gears B and C.



6. Discuss the stability of two-wheel vehicles. 3
7. A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg 6
- respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45° , B to C 70° and C to D 120° . The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.
8. Derive expressions of vibration of two degree of freedom under Undamped, damped and 3x3
- forced damped conditions.

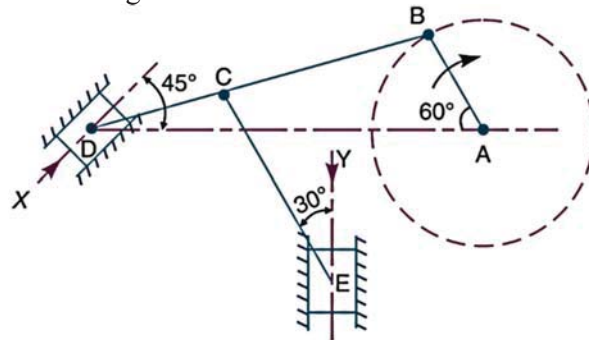
TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
(NEW COURSE)

Exam	Regular (Model Question)		
Level	B.E.	F. M.	60
Program	BAS	P. M.	24
Year / Part	II/II	Time	3 hrs.

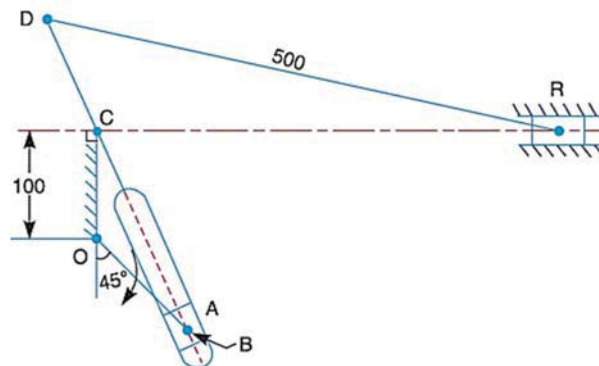
Subject: Theory of Machine (ENM 257)

- Candidates are required to give their answers in their own words as far as practicable.
- Attempt all questions. Figure in the margin indicates Full marks.
- Assume suitable data if necessary.

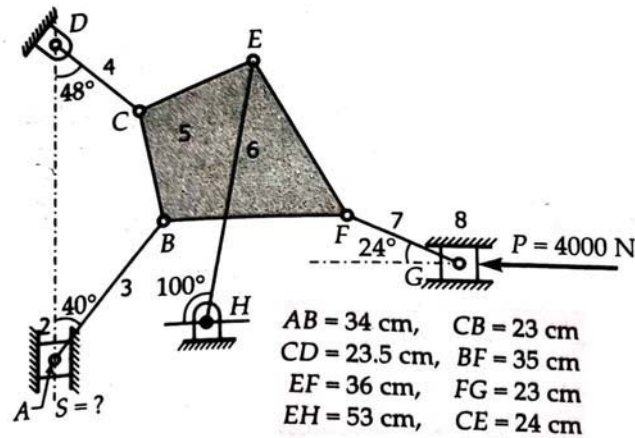
1. Discuss the method of position analysis using loop closure equation for four bar mechanism. Calculate the degree of freedom of mechanism shown in figure. [3+5]



2. Verify Aronhold Kennedy theorem in a mechanism. [4]
3. In a Whitworth quick return motion, as shown in Fig, OA is a crank rotating at 30 r.p.m. in a clockwise direction. The dimensions of various links are: OA = 150 mm; OC = 100 mm; CD = 125 mm; and DR = 500 mm. Determine the acceleration of the sliding block R and the angular acceleration of the slotted lever CA. [8]

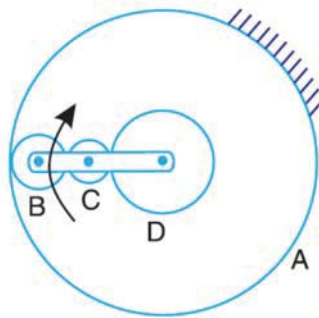


4. Discuss the method of complex number method in analysis of linkage force. [4]
5. For a riveter mechanism shown in figure determine the magnitude of S force on the slider 2 to maintain the mechanism in equilibrium. [8]



6. A cam is to give the following motion to a roller follower of 10mm radius: [8]
 1. Outstroke during 60° of cam rotation in a cycloidal motion of follower;
 2. Dwell for the next 120° of cam rotation;
 3. Return stroke during next 60° of cam rotation in a uniform motion, and
 4. Dwell for the remaining 120° of cam rotation.

The stroke of the follower is 50 mm and the minimum radius of the cam is 60 mm. Draw the profile of the cam when the axis of the follower is 20mm offset from the axis of the camshaft.
7. Explain the method of force calculation on spur gear teeth. [4]
8. Discuss on the requirement of minimum number of teeth on the pinion in order to avoid interference in rack and pinion gear system. [4]
9. Explain velocity ratio and center distance of worm and worm gear. [4]
10. An epicyclical gear train, as shown in Fig. is composed of a fixed annular wheel A having 150 teeth. The wheel A is meshing with wheel B which drives wheel D through an idle wheel C, D being concentric with A. The wheels B and C are carried on an arm which revolves clockwise at 100 r.p.m. about the axis of A and D. If the wheels B and D have 25 teeth and 40 teeth respectively, find the number of teeth on C and the speed and sense of rotation of C. [8]



Exam.	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BCT	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: - Operating System

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All the Questions.
- ✓ Figures in margin indicate full marks.
- ✓ Necessary figure is attached herewith.

1. Define Bootloader. Explain any 2 types of boot mechanism. How does an operating system act as Extended machine? Explain. [1+ 3+4]
2. What are the advantages of multithreading? For given process, draw a Gantt Chart and calculate average waiting time for: FCFS and Round Robin (with Quantum Time = 4). Their arrival time, service time in milliseconds. [2+5]

Processes	AT	BT
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

3. a) What is Petersons Solution? Explain solution of Reader-Writer problem using semaphore with its respective pseudo-code implementations. [2+5]
- b) Under what condition does the solution to the Dining philosopher using semaphore leads to a deadlock conditions? Consider a system with 5 processes (p0, p1, p2, p3, p4) and 4 resource types (R0, R1, R2, R3). The number of instances of each resource type in the systems are: 9,7,6,4 respectively. Allocation table and Maximum claim table are shown below. Is the state safe? If so, show the safe execution of the processes. [2+6]

Allocation

	R0	R1	R2	R3
P0	3	1	2	2
P1	2	2	1	1
P2	2	2	1	0
P3	1	0	1	0
P4	1	2	0	1

Maximum Need

	R0	R1	R2	R3
P0	4	3	2	2
P1	2	3	1	3
P2	2	2	4	2
P3	3	2	1	0
P4	2	1	0	1

4. a) Explain any two approaches that helps to achieve the concept of stable storage. An HDD Disk with capacity of 512 Giga Bytes and an SSD disk with capacity of 256 Giga Bytes

cost NPR. 3,500 and NPR 5,600 respectively. Now, calculate the cost of storing a 300 MB file on each disk. Which disk is cost efficient? [4+3]

b) Given references to the following pages by a program 1, 2, 3, 2, 1, 5, 2, 1, 6, 2, 5, 6, 3, 1, 3, 6, 1, 2, 4, 3. How many page faults will occur if the program has 4-page frames available to it and use LRU, FIFO and Optimal page replacement algorithm. [5]

5. Explain the advantages and disadvantages of a contiguous file allocation scheme? [3]
6. Explain private and public key cryptography with examples. [4]
7. Explain Hypervisors and its types with examples? [4]
8. Explain any one functions of OS you studied as case study with its contributions. [3]
9. Write short notes on (any 2): (2*2=4)
 - i. Dockers and Kubernetes
 - ii. Secure Boot
 - iii. Sandboxing
 - iv. WSL

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
(NEW COURSE)

Exam.	Regular (Model Question)		
Level	Bachelor	Full Marks	60
Program	BCE	Pass Marks	24
Year/ Part	II/II	Time	3hrs.

Probability & Statistics (ENSH 253)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Tables are attached herewith
- ✓ Assume suitable data if necessary.

- 1 The construction times (in days) for retaining walls by two contractor-A and contractor-B, recorded at 8 different project sites of each contractor and are provided in the table below:

	Site and Construction time (days)							
contractor	1	2	3	4	5	6	7	8
A	15.2	14.8	13.9	14.1	16.0	13.8	14.5	14.7
B	12.3	10.1	14.2	11.6	13.7	15.5	11.2	12.5

- a) Determine contractor shows more consistent construction time.
- b) For a time-sensitive project, which contractor would you recommend and why?
- 2 Define the multiplicative law of probability. The probability that a contractor will get a plumbing contract is $2/3$, and the probability that he will not get an electric contract is $5/9$. If the probability of getting at least one contract is $4/5$, what is the probability that he will get both the contracts?
- 3 Discuss difference between discrete random variable and continuous random variable. support you answer with suitable example.
- 4 Define the Poisson Distribution,
 During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 3. What is the probability that 4 particles enter the counter in a given millisecond?
- 5 Define standard normal distribution, mention it's area properties. A civil engineer is testing compressive strength of a certain type of concrete. Based on previous tests, the compressive strength of this concrete is normally distributed with mean of 30 MPa and a standard deviation of 4 MPa .
 What is the probability that a randomly selected sample of this concrete will have a compressive strength
- a) greater than 38 MPa
- b) between 40 and 45 MPa
- c) less than 25 MPa

OR

A population consists of the values: 2, 4, 6, 8, 10.

Consider all possible samples of size two, drawn without replacement from this population.

Find:

- (a) The sampling distribution of the sample mean
- (b) Show that the mean of the sampling distribution of the sample mean is an unbiased estimator of the

- population mean
(c) The standard error of the sampling distribution of the sample mean

- 6 Define estimation and properties of good estimators.
A random sample of 50 articles selected from a batch of 2000 articles shows that the average diameter of the article is 0.345 with standard deviation of 0.048. Find 95% and 98% confidence interval for the average of this batch of 2000 articles. 5

- 7 Discuss the hypothesis testing procedure of single population mean for small sample. 5

- 8 Three different machines are being considered for use in manufacturing rubber seals. The machines are being compared with respect to tensile strength of the product. A random sample of four seals from each machine is used to determine whether the mean tensile strength varies from machine to machine. The following are the tensile-strength measurements in kilo grams per square centimeter $\times 10^{-1}$

	tensile-strength				
Machine	1	17.5	16.9	15.8	18.6
	2	16.4	19.2	17.7	15.4
	3	20.3	15.7	17.8	18.9

5

Perform the analysis of variance at the 0.05 level of significance and indicate whether or not the mean tensile strengths differ significantly for the three machines.

- 9 A researcher is analyzing the impact of economic condition on the academic performance of students. For this study, 2,00 students were classified according to their intelligence level and economic condition as shown in the table below:

Economic Condition	Intelligence			Total
	Excellent	Good	Dull	
Good	13	46	40	99
Not Good	19	40	42	111
Total	32	86	82	200

5

Test whether there is a statistically significant association between intelligence level and economic condition at the 5% level of significance.

- 10 The following table gives age (yrs) and blood pressure(mm/hg) of 10 adults.

Age(x)	30	50	40	35	55	60	43	58	65	70
BP(y)	122	150	122	120	140	142	150	130	145	160

4+1

Find out if there is any correlation between age and BP. Also, find the coefficient of determination. Discuss the application of coefficient determination.

- 11 The thrust of an engine (y) is a function of temperature (x) in °F, assuming all other important variables are held constant. Consider the following data.

Temperature (x)	17	16	14	13	18	16	15	17	12
Thrust (y)	43	46	32	31	49	40	38	45	30

5

- Fit a simple linear regression to estimate thrust with respected to temperature.
- Develop the scatter plot of above data
- Study result of a) and b) discuss the relation between temperature and thrust of engine.

- 12 An engineering construction firm is monitoring the compressive strength of certain material using an $\bar{X}$ control chart. For this purpose, they collect samples ($n = 4$) at regular intervals over 10 different batches. The compressive strength (in MPa) of each sample in a batch is recorded as follows:

Batch	1	2	3	4	5	6	7	8	9	10
Mean($\bar{X}$)	31.5	32.3	30.0	31.9	33.8	30.6	32.2	30.0	31.0	32.7
Range	0.6	0	0.9	0.6	0.2	0.6	0.4	0	0	0.2

5

- 1) Construct the mean ($\bar{X}$) control chart and plot the mean of each batch
- 2) What is decision on function of machine.
- 3) Given $A_2 = 0.729$ for sample size $n = 4$

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
(NEW COURSE)

Examination	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BAM	Pass Marks	24
Year/ Part	II/II	Time	3 hrs.

Subject:

Signal and System

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **ALL** questions
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define CT and DT signals. Explain the energy and power signals. Determine whether the signal $x(t) = 5 \cos 5t$ is power or energy signal. [2+3+3]
2. Derive the necessary condition for the signal $x[n] = e^{j\omega n}$ to be periodic. [3]
3. Derive the Fourier series expansion of a continuous time periodic signal $x(t)$. [5]
4. State and prove Parseval's relation for Discrete Time Fourier Series. [5]
5. Differentiate Fourier series and Fourier transform? Find the response of the LTI system with impulse response $h(t) = e^{-at} u(t)$, $a > 0$ to the input signal $x(t) = e^{-bt} u(t)$, $b > 0$. [3+5]
6. Explain the duality and symmetry property of the Fourier Transform. Find the Fourier Transform of the unit step signal and the rectangular signal. [2+8]
7. Define LTI system. Explain causality and stability property of continuous time LTI system. [2+3]
8. Find the output of an LTI system, for input $x(t) = u(t+1) - u(t-1)$ and $h(t) = u(t+1) - u(t-1)$. [5]
9. State sampling theorem with mathematical expression. [4]
10. Explain transfer function. Evaluate the Step response of low pass filter. [2+5]

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EXAMINATION CONTROL DIVISION
(NEW COURSE)

Exam.	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BCE	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: - Soil Mechanics (CE552)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
 - ✓ Attempt **All** questions.
 - ✓ The figures in the margin indicate **Full Marks**.
 - ✓ Assume suitable data if necessary.
- Define soil and soil mechanics. Write down its importance. [2]
 - What are the various soil structure in the soil? Describe them. [6]
 - Draw the phase diagram for the saturated, partially saturated and dry soil. [3]
 - Enlist the index properties of Coarse-grained soil and fine- grained soil. [2]
 - A soil deposit has a void ratio of 0.9. If the void ratio is reduced to 0.6 by compaction, find the percentage of volume loss due to compaction? [3]
 - Classify the following soils a, and b as per IS soil classification system:
 - Soil 'a' passing from 75 μ sieve = 4%, soil passing from 4.75mm sieve (coarse fraction) = 62%, coefficient of uniformity = 5, coefficient of curvature = 2.6
 - Soil 'b' passing from 75 μ sieve = 62%, liquid limit = 54%, plastic limit = 23% [2+2]
 - A permeameter of diameter 82.5 mm contains a column of fine sand 460 mm long. When water flows through it under constant head at a rate of 191 ml/min, the loss of head between two points 250 mm apart is 380 mm. Calculate coefficient of permeability. If falling head test is made on same sample a sand pipe of diameter 30 mm, in what time the water level in sand pipe will fall from 1560 mm to 1060 mm? (3+2)
 - Derive Laplace equation and prove that flow lines and equipotential lines are orthogonal. (3)
 - A sand deposit consists of two layers. The top layer is 3.0m thick ($\gamma = 17\text{KN/m}^3$) and bottom layer is 4.0m thick ($\gamma_{\text{sat}} = 21\text{KN/m}^3$). The water table is at a depth of 4.0m from the surface. Draw the diagrams, showing the variation of total stress, neutral stress and effective stress.
 - Describe approximate stress distribution methods for loaded areas. [6+2]
 - Differentiate between normally consolidated and over consolidated Clay. A compressible layer whose total settlement under a given loading is expected to 20 cm settles 4cm at the end of 2 months. How many months will be required to reach a settlement of 10 cm? Assume double drainage. [2+6]
 - Explain the Critical State Framework in soil mechanics. How does it help in understanding the shear strength behavior of soils?
 - Draw the Mohr–Coulomb strength envelope for cohesive soil, cohesion less soil and purely cohesive soil.
 - A sample of dry cohesion less soil was tested in triaxial machine. If the angle of shearing resistance was 36° and the confining pressure, 100 kN/m^2 , determine the deviator stress at which the sample failed. (3+3+6)

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
(NEW COURSE)

Exam.	Regular (Model question)		
Level	B. E.	Full Marks	60
Programme	B.Arch.	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

Subject: - Structure II (ENCE260)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
 - ✓ Attempt **All** questions.
 - ✓ The figures in the margin indicate **Full Marks**.
 - ✓ Assume suitable data if necessary.
 - ✓ Use of IS codes: IS 800(2007) and IS456:2000(2021) are allowed.
-

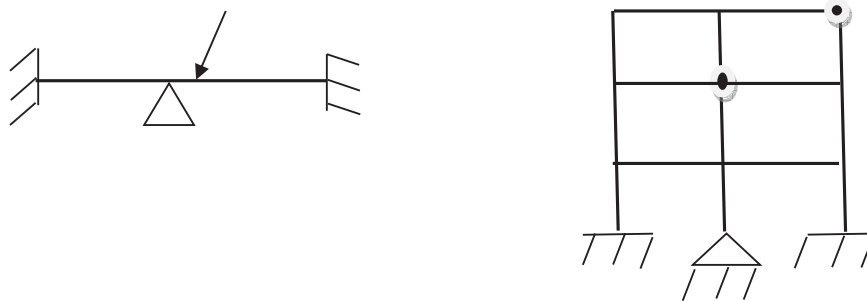
1. What is the structural use of timber? Explain with neat sketches. Calculate the load carrying capacity of a circular timber column of radius 180mm. The compressive stress parallel to the grain is 10.6MPa and Young's modulus = 12,000 MPa. [2+6]
2. (a) Explain the modes of out-of-plane failure of masonry structures. [4]
(b) What are the mechanical properties of Structural steel? How can you measure it? [4+2]
3. (a) Design fillet welded connection using 10mm gusset plate to connect channel section MC200 to carry service axial load of 250kN. Assume welding is done at workshop. [6]
(b) Design a simply supported rectangular RCC beam having 5 m effective span which is subjected to factored live load of 14 kN/m. Consider the self-weight of beam as dead load. Use M20 concrete and Fe500 rebar. [6]
4. (a) Design interior panel of RCC slab for a room measuring 3 m x 5 m is subjected to floor finish load of 1.5 kN/m² and Live load of 5kN/m². The slab is supported on 350mm wide beam on all sides. Use M25concrete and Fe 500 TMT rebar. Check for deflection is required. [10]
(b) Explain how the distribution of the mass and stiffness affect the seismic behavior of the building. [5]
5. Write short notes on following: [3*5]
 - a) Plan irregularity of building
 - b) Limit state of serviceability
 - c) Strong Column and weak beam theory
 - d) Stiffness of Structure
 - e) Development length of reinforcing bar in reinforced concrete member

Exam.	Regular (Model Question)		
Level	BE	Full Marks	60
Programme	BAG	Pass Marks	24
Year / Part	II / II	Time	3 hrs.

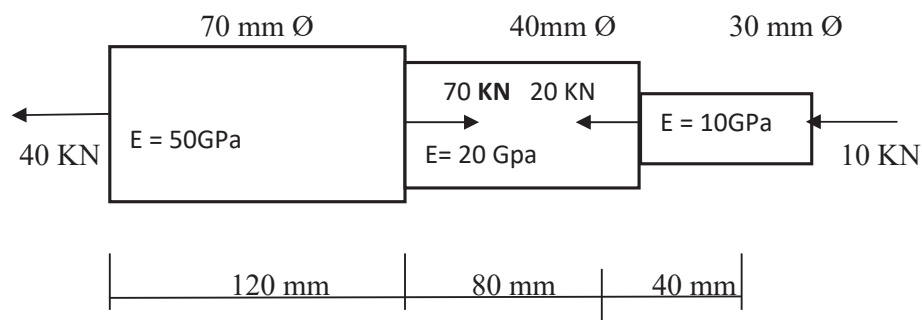
Subject: - Strength of Material and Theory of Structure (ENCE 257)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All the Questions.
- ✓ Figures in margin indicate full marks.
- ✓ Necessary figure is attached herewith.

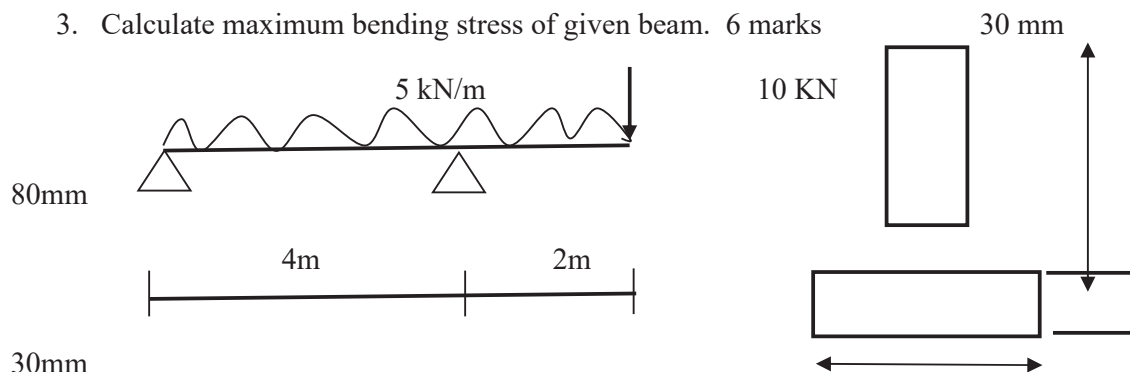
1. Calculate static and kinematic indeterminacy of given structure. 5 marks



2. Calculate total deformation of given composite bar. 7 marks



3. Calculate maximum bending stress of given beam. 6 marks



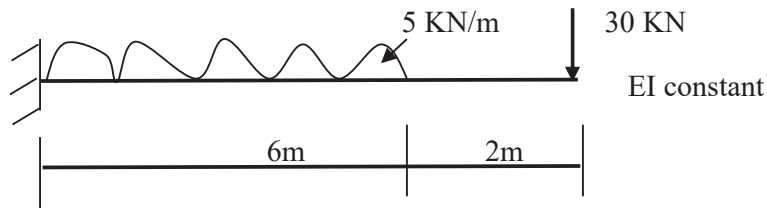
4. Find the buckling strength of column of length 8 m and x section is 80 mm

20 cm X 30 cm

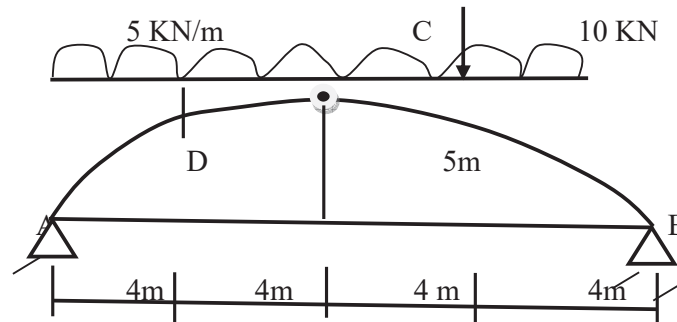
6 marks

- a) one end fixed another end free
- b) one end fixed another end hinged.FOS = 2.5

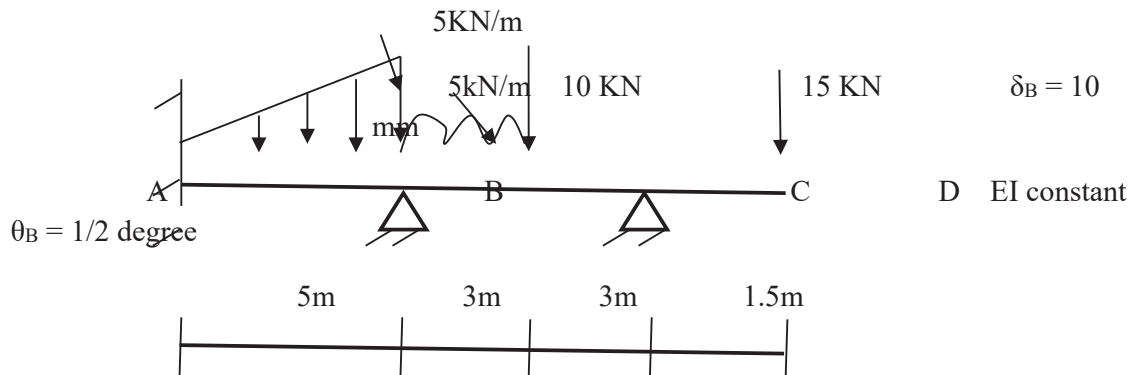
5. Derive expression of slope and deflection and also find slope and deflection at free end. 8 marks



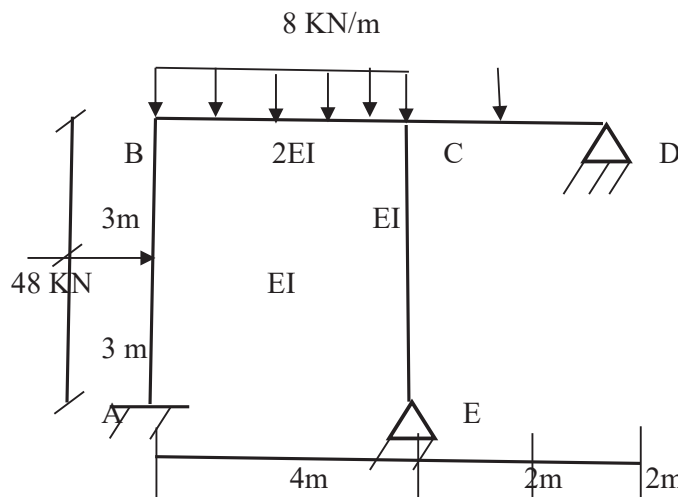
6. Define ILD and explain its uses. 4 marks
7. Find out BM, radial shear and Normal Thrust at D of given parabolic arch. 8 marks



8. Analyze the given beam using slope deflection method. 10 marks



9. Analyze the given frame using moment distribution method. 10 marks



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
(NEW COURSE)**

Exam	Regular (MODEL QUESTION)		
Level	BE	Full Marks	60
Programme	BAG	Pass Marks	24
Year/Part	II/II	Time	3 hrs.

.....
SUBJECT: - TRACTOR SYSTEM AND CONTROL
.....

- ✓ Attempt all questions
- ✓ All questions carry equal marks

1. How tractors are classified? List out them and explain in detail.
2. What are brake boosters and how do they work? Classify them and describe their constructions.
3. Explain the role of tyre pressure and how does pressure affect the tyre? Why liquid ballasting is done in tractor tyre? Explain the liquid ballasting procedure.
4. How tractor differential works? Explain with neat diagram and why differential lock is used in tractor and not in other vehicles?
5. What parameters do you consider during the tractor design? Explain in detail.
6. How tractor's hydraulic system works? Explain the 4/3 types of direction control valve.
7. Explain the single drop arm steering system with the help of neat diagram and also further describe the function and construction of its components.
8. A Swaraj 735 tractor has 350 mm radius of its wheel and engine rated rpm is 2,000 and the wheel speed is 200 rpm; now determine the theoretical speed of Tractor and distance covered by tractor in 1 hour? Similarly, travel reduction ratio of the tractor is 20 % then what will be the actual speed of the tractor?
9. How power is tapped from the tractor to operate agricultural machineries? List out the types of PTO and explain about independent PTO.
10. A tractor having engine power 61.1 kW and rear wheel weight was measured 3900 kg was practically tested on actual field and following data was measured. Tractor was operated 55.8 meters without load and again operated with load 46.2 meters in 25.8 seconds and drawbar pull was measured 26.2 kN and at the same time fuel was consumed was 176 grams. Find: Slip of the wheel, travel speed km/hr, drawbar power, specific fuel consumption and wheel power assume transmission efficiency.

TRIBHUWAN UNIVERSITY
INSTITUTE OF ENGINEERING
EXAMINATION CONTROL DIVISION
(NEW COURSE)

Exam.	Regular (Model Question)		
Level	BE	Full marks	60
Programme	BCE	Pass marks	24
Year /Part	II/II	Time	3Hrs.

Subject: Water Supply Engineering (ENCE254)

- ✓ *Candidates must give their answers in their own words as far as practicable.*
- ✓ *The figures in the margin indicate **Full Marks***
- ✓ *Assume suitable data if necessary.*

Group A (7 x 2 = 14)

1. List the importance of water from WASH perspectives.
2. Define alternative sources of water and provide examples.
3. Justify the use of peak factors in transmission and distribution mains.
4. Highlight the engineering and environmental significance of turbidity.
5. Define a membrane filter and list its types with the impurities each type removes.
6. What is meant by smart water distribution system and list its key features.
7. Recommend the practicability of socket and electro-fusion pipe joints.

Group B (7 x 4 = 28)

1. Justify the key criteria for selecting a safe and reliable water source for village water supply.
2. A proposed spring has a safe yield of 5 L/s. If the per capita water demand is 65 L/capita/day, calculate the current population that can be served under the scheme, assuming a design period of 20 years and an annual population growth rate of 1.7%.
3. The chemical analysis of a water sample yields the following ion concentrations in mg/L:
 $\text{Na}^+=20$, $\text{K}^+=15$, $\text{Ca}^{++}=25$, $\text{Mg}^{++}=12$, $\text{Cl}^- = 40$, $\text{HCO}_3^- = 60$, $\text{SO}_4^{--}=15$, $\text{NO}_3^- = 8$; Calculate the total hardness, carbonate hardness, and non-carbonate hardness in mg/L as CaCO_3 .
4. Draw a neat, labeled sketch of a spring intake structure, highlighting its main components and their functions.
5. Define “destabilizing colloids” in the context of water treatment. Justify the use of this technique in water treatment plants?
6. Describe the working mechanisms of desalination and reverse osmosis process, along with their suitable applications.
7. Define Interruption Chamber (IC) and Break Pressure Tank (BPT) and justify their important role in particular locations of water supply system.

Group C (3x6 =18)

1. A rectangular sedimentation tank is designed to treat 10 MLD of water. A detention basin with a width-to-length ratio of 1:3 is proposed to trap all particles larger than 0.04 mm. Assuming the particle specific gravity at 20°C is 2.65, compute the tank dimensions. If the tank depth is 3.5 m, calculate the detention time.
2. An experiment shows that a concentration of 0.1 g/m³ of free available chlorine achieves a 2-log reduction of bacteria in 8 minutes. What contact time is required to achieve a 3-log reduction of bacteria at a free available chlorine concentration of 0.05 g/m³? Assume Chick's Law and Watson's Law apply with $n = 1$.
3. Design the pipes RA and AB for the water distribution network shown below. Assume a per capita water demand of 150 L/capita/day with a peak factor of 3, and use a Hazen-Williams coefficient of $C = 100$. Ensure that the residual pressure at any point in the system is at least 10 m. Determine the required pipe diameters for RA and AB and verify that the velocities in the pipes are within acceptable limits. Clearly state all assumptions used in your design and show all calculations.

